PROBLEM 2 | Problems T2_CrystStruct.pdf

Crystalline Structures & Miller Indices

Atomic Mass and Density of FCC Magnesium

Original statement · Problems T2_CrystStruct.pdf

> *2. Determine the atomic mass of a metal element with FCC structure, density of* $1.74\ \mathrm{g/cm^3}$ *and lattice constant of* $4.527\ \text{Å}$*. Find the element with these characteristics.* *(Solution:* $M = 24.3\ \mathrm{g/mol}$*; Mg)*
>
> **Solution:** $M = 24.3\ \mathrm{g/mol}$; Mg

PHASE 1: HYPOTHESIS & PARAMETERS

Input Data:

- Crystal structure: Face-Centred Cubic (FCC) $\implies n = 4$ atoms/cell.
- Lattice parameter: $a = 4.527\ \text{Å} = 4.527 \times 10^{-8}\ \mathrm{cm} = 0.4527\ \mathrm{nm}$.
- Macroscopic volumetric density: $\rho = 1.74\ \mathrm{g/cm^3} = 1740\ \mathrm{kg/m^3}$.
- Avogadro constant: $N_A = 6.022 \times 10^{23}$ atoms/mol.

Starting Hypotheses:

1. The metal is a pure single crystal free of vacancies or defects that would significantly alter the apparent density.
2. The cubic unit cell satisfies $V_C = a^3$.

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

Citing the theoretical foundations of the course [Session 4 Slide 7]:

- The theoretical volumetric density of a crystalline solid is defined by the ratio of the cell mass to its volume:

$$\rho = \frac{m_{\text{cell}}}{V_C} = \frac{n \cdot M}{a^3 \cdot N_A}$$

- Solving algebraically for the molar atomic mass M:

$$M = \frac{\rho \cdot a^3 \cdot N_A}{n}$$

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

1. **Unit cell volume (V_C):**

$$a = 4.527 \times 10^{-8}\ \text{cm}$$

$$V_C = a^3 = (4.527 \times 10^{-8}\ \text{cm})^3 = 9.2775 \times 10^{-23}\ \text{cm}^3$$

2. **Molar mass (M):**

$$M = \frac{(1.74\ \text{g/cm}^3) \times (9.2775 \times 10^{-23}\ \text{cm}^3) \times (6.02214 \times 10^{23}\ \text{mol}^{-1})}{4}$$

$$M = \frac{1.6143 \times 10^{-22} \times 6.02214 \times 10^{23}}{4} = \frac{97.215}{4} = \mathbf{24.304}\ \text{g/mol}$$

3. **Identification in the Periodic Table:**

Checking the standard atomic weights of the elements:

- Sodium (Na): 22.99 g/mol
- **Magnesium (Mg):** 24.305 g/mol (our 24.304 g/mol agrees to 0.004%; the official key rounds to 24.3).
- Aluminium (Al): 26.98 g/mol

Therefore, the element with these physical characteristics and atomic mass is **Magnesium (Mg)**.

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

- **Crystallographic Analysis:** At ambient temperature and pressure, pure magnesium usually crystallises in the **HCP** structure ($c/a \approx 1.624$) with density $1.738\ \text{g/cm}^3$. However, in thin films, epitaxial metastable states or as a theoretical polymorphic phase, it can adopt the FCC cubic configuration while keeping the same bulk atomic density.
- **Dimensional Consistency:**

$$[M] = \frac{[\mathrm{g/cm^3}] \cdot [\mathrm{cm^3}] \cdot [\mathrm{mol^{-1}}]}{[\mathrm{dimensionless}]} = \mathrm{g/mol} \quad \checkmark$$